

PROJECT SUBMISSION REPORT

Course Code: CSC-483

Course Name: Data Science

Project Title: Pakistan Used Car Price Prediction and Market Analysis System

Student Name: Maaz Alvi

Student ID: IU02-0322-0038

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1. Introduction

The automobile industry plays a significant role in Pakistan, and the demand for used vehicles continues to grow every year. Determining the fair market value of a used car can be difficult because prices depend on multiple factors such as brand, model, fuel type, mileage, manufacturing year, and city of registration. This project focuses on analyzing used car listings from OLX Pakistan and developing a machine learning model capable of predicting vehicle prices. The system helps users estimate a reasonable price based on the characteristics of a vehicle while also providing valuable insights into the Pakistani used-car market.

2. Dataset Description

The dataset was collected from OLX Pakistan and contains information about used vehicles listed for sale. The dataset includes both numerical and categorical attributes that influence vehicle pricing.

Attribute	Description
Brand	Vehicle manufacturers such as Suzuki, Toyota, Honda
Condition	Vehicle condition (Used/New)
Fuel	Fuel type such as Petrol, Diesel, CNG
KMs Driven	Total mileage driven by the vehicle
Model	Specific vehicle model
Registered City	City where the vehicle is registered
Transaction Type	Cash or Installment
Year	Manufacturing year of the vehicle
Price	Target variable representing vehicle price

- Analyze the Pakistan used-car market using OLX data.
- Clean and preprocess the dataset to improve data quality.
- Perform Exploratory Data Analysis (EDA) to identify trends and patterns.
- Determine the most important factors affecting car prices.
- Build a machine learning model for price prediction.
- Develop a simple prediction system that estimates vehicle prices based on user input.

3. Methodology

The project followed a complete machine learning workflow for developing a used car price prediction system. The process began with data profiling to understand the structure, quality, and characteristics of the dataset. After profiling, several data preprocessing techniques were applied to improve data quality and prepare the data for machine learning.

Preprocessing included:

- Missing Value Handling: Missing values were identified and appropriately treated.
- Duplicate Removal: Duplicate records were removed to reduce bias and improve reliability.
- Outlier Detection and Removal: Extreme values were eliminated to enhance model performance.
- Label Encoding: Categorical features such as Brand, Fuel, Model, City, and Transaction Type were converted into numerical values.

Following preprocessing, Exploratory Data Analysis (EDA) was conducted to identify patterns and relationships among variables. Visualizations and statistical analysis were used to examine vehicle pricing behavior. A correlation heatmap revealed:

- Price and Year: Positive correlation (0.54)
- Price and KMs Driven: Weak negative correlation (-0.006)
- Year and KMs Driven: Negative correlation (-0.19)

The analysis indicated that newer vehicles generally have higher market prices, while mileage alone has a relatively limited effect on price.

To further understand model behavior, feature importance analysis was performed using the Random Forest model. The most influential features were:

- Manufacturing Year: 38.6%
- Brand: 26.0%
- Model: 14.7%

The dataset was then divided into training and testing sets, and a Random Forest Regressor was trained due to its effectiveness in handling both numerical and categorical data while maintaining strong predictive performance. Model evaluation produced an R^2 Score of

0.835, indicating that approximately 83.5% of the variation in vehicle prices could be explained by the model.

A prediction system was finally developed using the trained model to estimate vehicle prices based on user-provided vehicle characteristics.

4. Environmental Setup

The project environment was configured to support data processing, analysis, model training, and prediction generation.

The implementation environment included:

- Programming Language: Python
- Development Environment: Jupyter Notebook / Google Colab (*depending on project environment*)
- Libraries Used:
 - Pandas – data manipulation and preprocessing
 - NumPy – numerical computations
 - Matplotlib & Seaborn – data visualization and EDA
 - Scikit-learn – machine learning model development and evaluation

The machine learning workflow involved:

1. Loading and profiling the dataset
2. Applying preprocessing techniques
3. Performing exploratory data analysis
4. Encoding categorical variables
5. Training the Random Forest Regressor
6. Evaluating model performance using R^2 score
7. Deploying the prediction mechanism for price estimation

A demonstration of the prediction system produced the following sample result:

Sample Input:

- Brand = Suzuki
- Model = Alto
- Fuel = Petrol
- Year = 2015
- KMs Driven = 50,000
- Registered City = Karachi
- Transaction Type = Cash
- Condition = Used

Predicted Output:

- Predicted Price = PKR 971,845

This demonstrates the practical use of the developed system for estimating used vehicle prices.

5. Conclusion

This project successfully developed a machine learning-based used car price prediction system for Pakistan. The work covered the complete machine learning pipeline, including data preprocessing, exploratory data analysis, feature importance analysis, model training, and prediction generation.

The Random Forest Regressor achieved an R^2 score of 0.835, demonstrating strong predictive capability and confirming that selected features effectively represent vehicle pricing behavior.

The developed prediction system can support buyers and sellers by providing estimated market prices based on important vehicle characteristics. The project highlights the practical application of data science and machine learning in solving real-world business problems.

For future enhancement, the system can be improved by integrating real-time vehicle listings, expanding dataset size, developing a web or mobile application, applying advanced machine learning techniques, and incorporating image-based vehicle condition analysis.